

Ph.D. Exam in Measure Theory and Analysis
May, 1991

Be sure to give complete proofs and justify your steps. Do not leave gaps.

1. State:
 - (a) The Lebesgue Dominated Convergence Theorem
 - (b) Fatou's Lemma
 - (c) The Radon-Nikodym Theorem
 - (d) $(L^1(S, \Sigma, \mu))^* = L^\infty(S, \Sigma, \mu)$
2. State and prove the Vitali Convergence Theorem for integrals (convergence in measure form).
3. Show that a subset A of the reals is a set of Lebesgue measure zero if and only if there exists a sequence of intervals I_m such that
 - (a) $\sum_m \text{length}(I_m) < \infty$.
 - (b) $A \subset \bigcup_{m=k}^{\infty} I_m$ for every k .
4. Let f be Lebesgue integrable on $\mathbb{R}$. Suppose that $\int_0^x f dm = 0$ for every $x \in \mathbb{R}$. What can you conclude about f ?
5. State and prove Fubini's Theorem.
6. Let $(S_m, \Sigma_m, P_m)_{m=1}^{\infty}$ be a sequence of probability spaces. Let $E_m \in \Sigma_m$, $m = 1, 2, \dots$.
 - (a) Show that $E = \bigcap_{m=1}^{\infty} E_m \in \bigotimes_{n=1}^{\infty} \Sigma_m$.
 - (b) Prove that $\left(\bigotimes_{m=1}^{\infty} P_m \right) (E) = \prod_{m=1}^{\infty} P_m(E_m)$